

Basel Problem

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1 Basel Problem

1.1 $\zeta(2) = \sum_{n=1}^{\infty} \frac{1}{n^2}$

Problem 1. Calculate (or prove)

$$\zeta(2) = \sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}.$$

Solution. Method 1. Double Integral.

Consider the double integral $I = \iint_{D_1} \frac{1}{1-xy} dx dy$, where $D_1 = \{(x, y) : 0 < x < 1, 0 < y < 1\}$.

$$\begin{aligned} \iint_{D_1} \frac{dx dy}{1-xy} &= \text{/Generalized power series multiplication theorem/} \\ &= \iint_{D_1} \sum_{n=0}^{\infty} (xy)^n dx dy = \int_0^1 \left(\int_0^1 \sum_{n=0}^{\infty} (xy)^n dx \right) dy \\ &= \int_0^1 \sum_{n=1}^{\infty} \frac{y^{n-1}}{n} dy = \sum_{n=1}^{\infty} \frac{1}{n^2}. \end{aligned}$$

/Transformation of coordinates or variable substitution/

$$\begin{aligned} \left[x = \frac{u-v}{\sqrt{2}}, y = \frac{u+v}{\sqrt{2}}, \frac{D(x,y)}{D(u,v)} = \begin{vmatrix} x'_u & x'_v \\ y'_u & y'_v \end{vmatrix} \right. \\ \left. \Leftrightarrow (u, v)^T = \begin{pmatrix} \cos -\frac{\pi}{4} & -\sin -\frac{\pi}{4} \\ \sin -\frac{\pi}{4} & \cos -\frac{\pi}{4} \end{pmatrix} (x, y)^T \right] \end{aligned}$$

$$\begin{aligned} I &= \iint_{D_2} \frac{1}{1 - \frac{u^2-v^2}{2}} \cdot 1 du dv \\ &= 2 \left[\iint_{D_{21}} \frac{1}{2 - u^2 + v^2} du dv + \iint_{D_{22}} \frac{1}{2 - u^2 + v^2} du dv \right] \\ &= 4 \left[\int_0^{\frac{\sqrt{2}}{2}} \left(\int_0^u \frac{dv}{2 - u^2 + v^2} \right) du + \int_{\frac{\sqrt{2}}{2}}^{\sqrt{2}} \left(\int_0^{\sqrt{2}-u} \frac{dv}{2 - u^2 + v^2} \right) du \right] \\ &= [u \triangleq \sqrt{2} \sin \theta, v \triangleq \sqrt{2} \cos \theta] \\ &= \frac{\pi^2}{18} + \frac{\pi^2}{9} = \frac{\pi^2}{6}. \end{aligned}$$

$$\text{i.e. } \zeta(2) = \sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}.$$